Metrics Match Consequences

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Recall and choose your starting point

> Record your starting explanation for Metrics Match Consequences before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Train, Validation, Test

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Train a Regression Model

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Regression and Residuals

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Metrics Match Consequences trace and record the first step you can explain without help.

Use W2: Count the **consequence**s for **Threshold**s and **consequence**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Metrics Match Consequences explanation against the supplied facts before discussing it.

Use the worked case for **Threshold**s and **consequence**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Scores/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare **threshold**s .5 and .2 using the same cost.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Metrics Match Consequences target to a trace and a justified conclusion.

> A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What **evidence** is needed before choosing a **threshold**?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Metrics Match Consequences case and show the deciding evidence.

Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

Connect metrics to **consequence**s.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Missing a diseased plant is considered more costly than a **false alarm**, but staff can review only ten alerts daily. Which evaluation trade-off should be discussed?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 25 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Metrics Match Consequences adds to Responsible Plant Classifier and keep the evidence.

Approve the Plant Classifier **metric suite**: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What **evidence** is needed before choosing a **threshold**?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Apply the boundary

Apply the boundary. Apply the supplied Metrics Match Consequences facts and keep your reasoning.

Which IDs are positive at .5 and .3?

My reasoning and evidence

W2 Count the consequences

> Count the consequences. Apply the supplied Metrics Match Consequences facts and keep your reasoning.

Give TP, FP, FN, TN at both **threshold**s.

My reasoning and evidence

W3 Use the declared cost

> Use the declared cost. Apply the supplied Metrics Match Consequences facts and keep your reasoning.

Calculate costs with FP cost 1 and FN cost 4.

My reasoning and evidence

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Metrics Match Consequences facts and keep your reasoning.

Rules supplied for this independent case

Cost and decision rules: blocked is positive; predict blocked when score is at least the threshold, including equality. A false positive costs 1 unit and a false negative costs 4 units. Total cost = FP + 4 × FN. Compare both thresholds on exactly these four records.

Scores/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare **threshold**s .5 and .2 using the same cost.

My reasoning and evidence

W5 Protect the final test

> Protect the final test. Apply the supplied Metrics Match Consequences facts and keep your reasoning.

Why not select the **threshold** with the lowest final-test cost?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Metrics Match Consequences

> Translate a threshold choice into missed-leaf and false-alarm costs. Explain the labelled step

> before checking the changed condition.

Translate a **threshold** choice into missed-leaf and false-alarm costs.

Scores/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at **score**>=**threshold**. Miss cost5; false-alarm cost1.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/25

Metrics Match Consequences

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Threshold0.7

TP 1, FN 1, FP 0, TN 2: cost5.





2. Threshold0.4

TP 2, FN 0, FP 1, TN 1: cost1.





3. Choose for stated costs

Threshold0.4 has smaller validation cost on these four cases.

CHANGE ONE THING

Miss and false-alarm costs swap to1 and 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Translate a **threshold** choice into missed-leaf and false-alarm costs.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Threshold**0.4 has smaller validation cost on these four cases.

1. **Threshold**0.4

2. Choose for stated costs

3. **Threshold**0.7

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Choose for stated costs **Threshold**0.4 has smaller validation cost on these four cases. Read the picture in order.

Threshold0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Miss and false-alarm costs swap to1 and 5. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches. Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

Repair a missing building block

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Regression Model (ai-4/22). Goal: Fit a line from a declared **training** candidate set. Short practice: May day3's evaluation label choose the candidate line?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

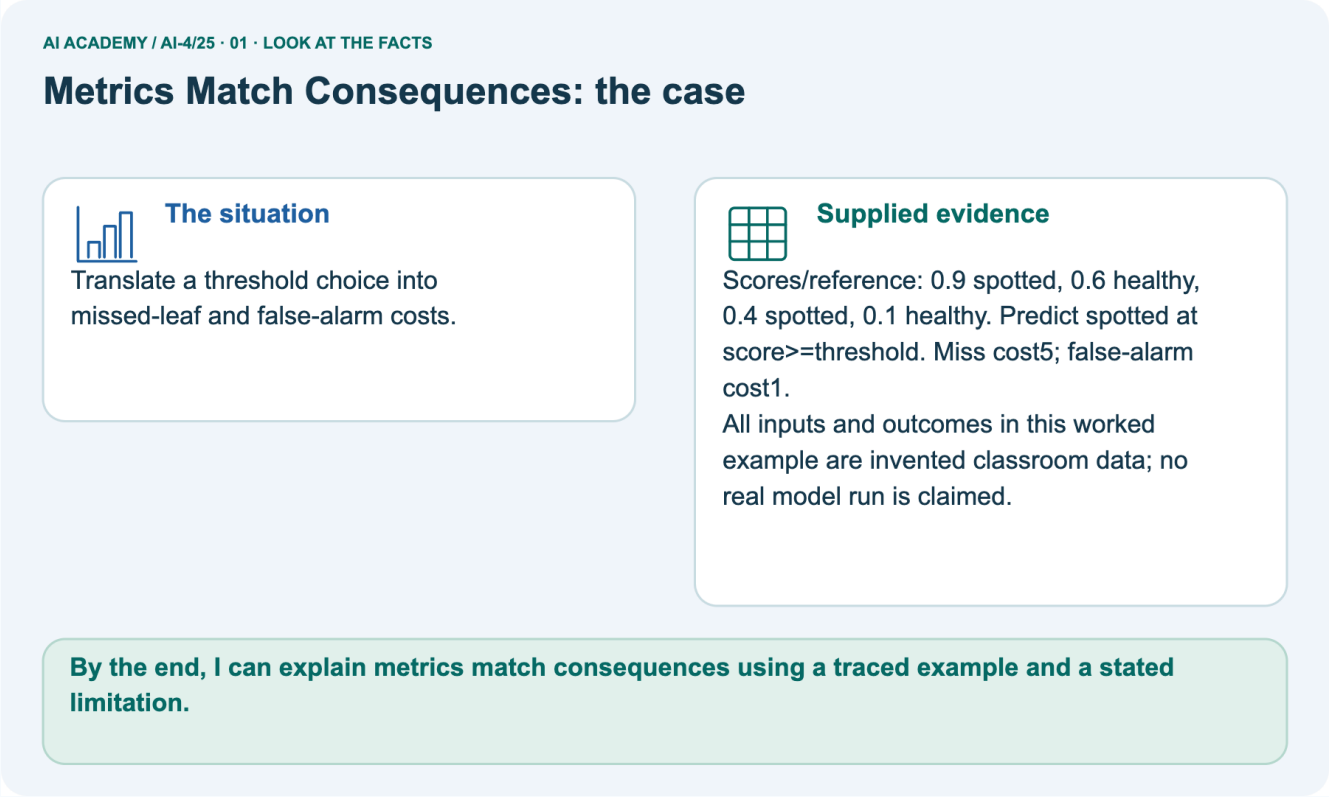
What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Metrics Match Consequences picture from its labels, then record a relevant limit.

01. Metrics Match Consequences: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

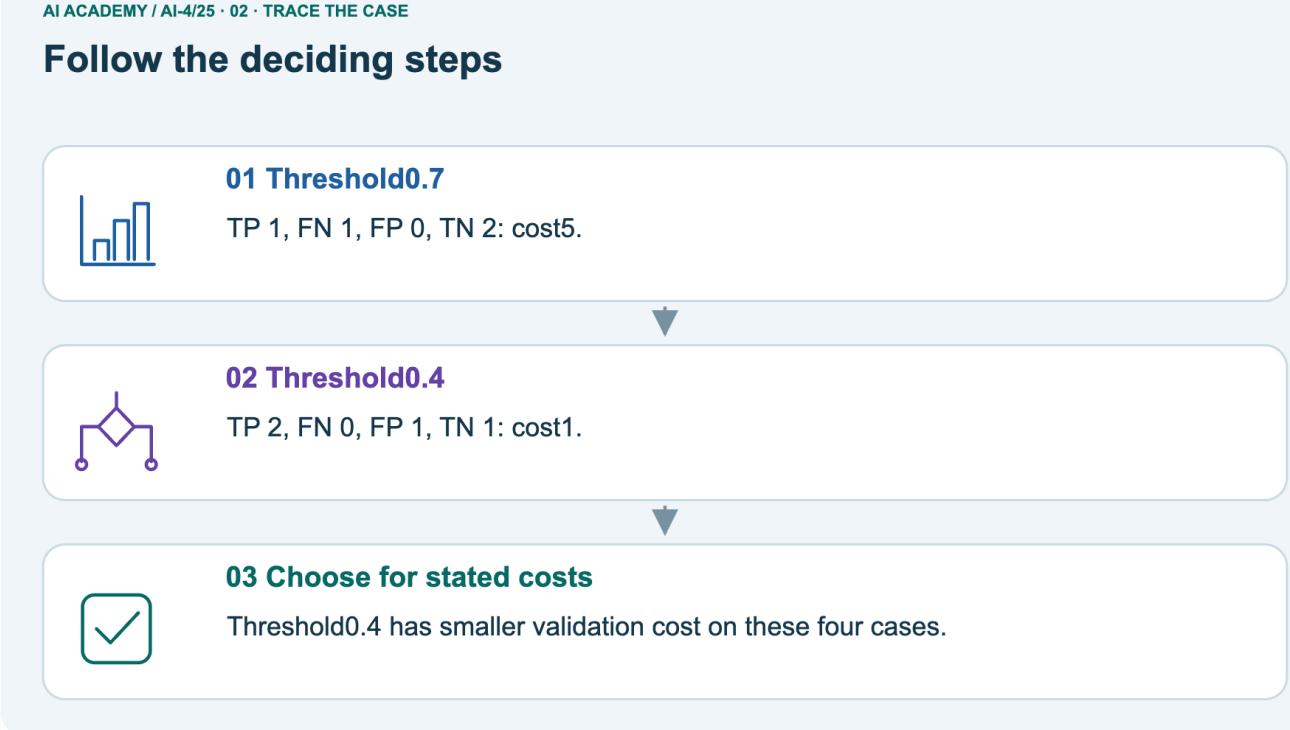
Translate a **threshold** choice into missed-leaf and false-alarm costs. Supplied facts: **Score**s/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at **score**>=**threshold**. Miss cost5; false-alarm cost1.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Threshold0.7: TP 1, FN 1, FP 0, TN 2: cost5. → **Threshold**0.4: TP 2, FN 0, FP 1, TN 1: cost1. → Choose for stated costs: **Threshold**0.4 has smaller validation cost on these four cases.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Thresholds and consequences at a glance

AI ACADEMY / AI-4/25 · 03 · INSPECT THE LESSON MODEL

Thresholds and consequences at a glance

Part	Evidence or action	Interpretation
Threshold .5	One false alarm one miss	Four of six correct
Threshold .3	Two false alarms no misses	Four of six correct
Decision rule	Include equality	Record before evaluation

Lowering the threshold here finds D but adds E as a false alarm. The two accuracies are equal while consequences differ. Precision is 2/3 at .5 and 3/5 at .3; recall rises from 2/3 to 1. This behavior follows these fixed scores and records, not a promise of future performance.

Follow the labelled picture with the matching explanation. Open the vector diagram

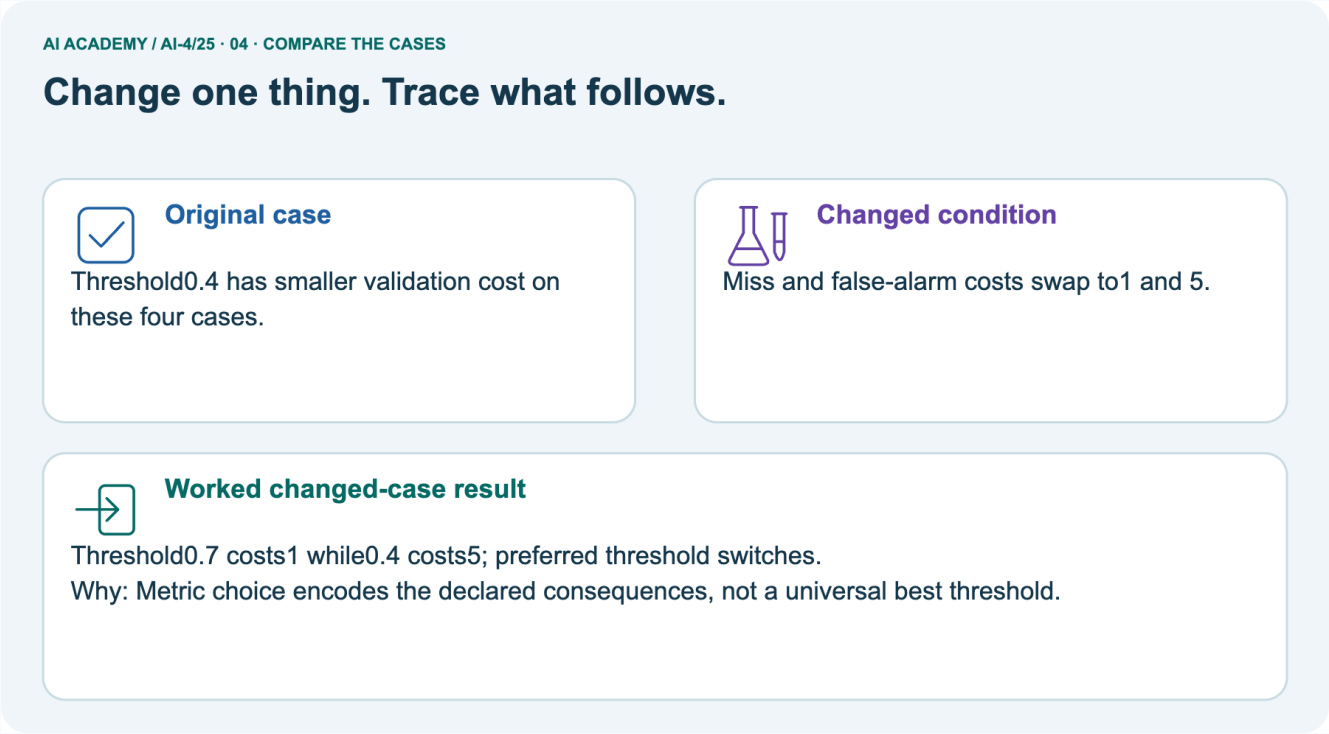
Thresholds and **consequence**s at a glance. **Threshold** .5 | One **false alarm** one miss | Four of six correct. **Threshold** .3 | Two **false alarm**s no misses | Four of six correct.

Decision rule | Include equality | Record before evaluation

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

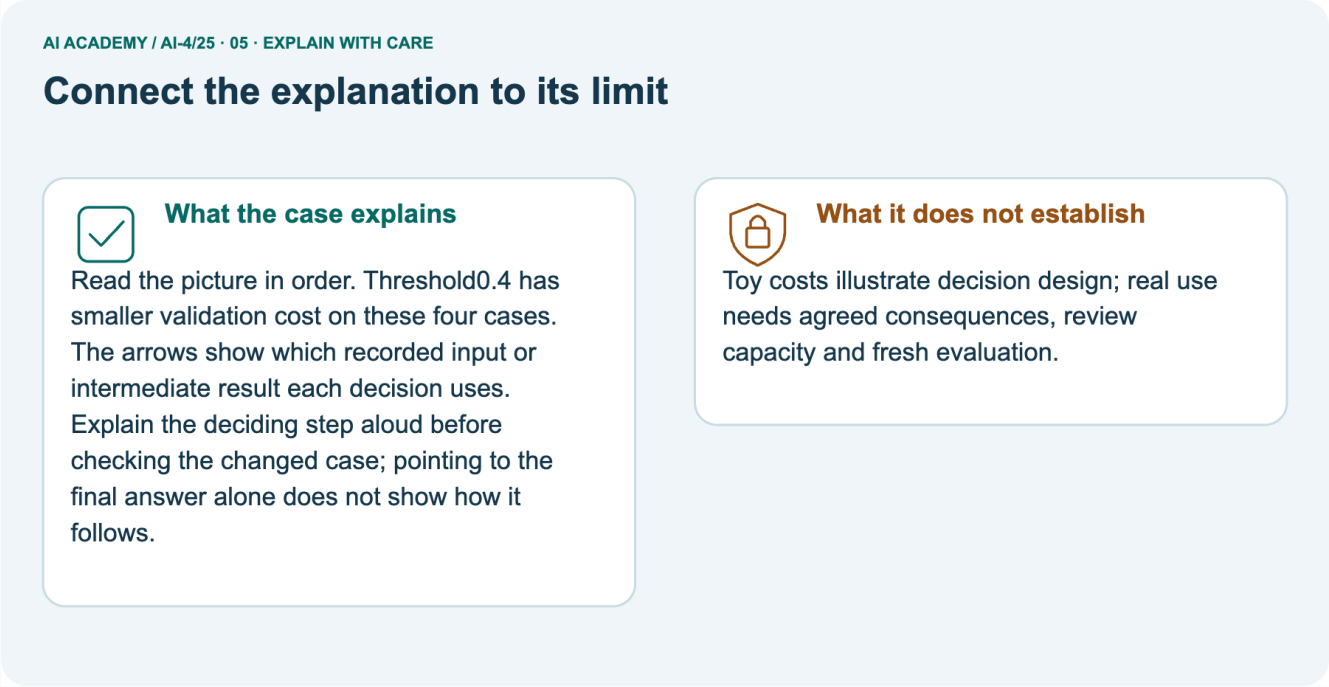
Original case: **Threshold**0.4 has smaller validation cost on these four cases. Changed condition: Miss and false-alarm costs swap to1 and 5. Expected result: **Threshold**0.7 costs1 while0.4 costs5; preferred **threshold** switches.

Reason: Metric choice encodes the declared **consequence**s, not a universal best **threshold**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Threshold**0.4 has smaller validation cost on these four cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Toy costs illustrate decision design; real use needs agreed **consequence**s, review capacity and fresh evaluation.

> Explain the case, then point to a limit on the claim.

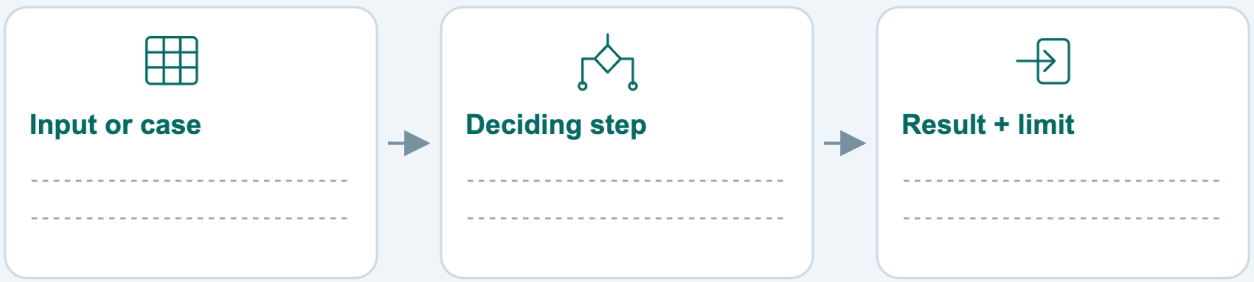
My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project

AI ACADEMY / AI-4/25 · 06 · DRAW AND EXPLAIN

Make the idea visible in your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Which labels determine the false-alarm count at threshold 0.4? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which labels determine the false-alarm count at **threshold** 0.4? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.